Generative Classifiers as Trustworthy Alternative in Tactile Classification

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I. Objective

Recent advancements in deep learning enable robots to perform complex tasks with human-like proficiency, sometimes exceeding human capabilities. Tulbure and Bäuml [1] demonstrate that high-accuracy material classification is achievable using a tactile deep learning network and commercially available tactile skin. However, state-of-the-art discriminative classifiers (DC) can show overconfidence with out-of-distribution (OoD) data, leading to occasional but significant "false positives." Due to their abstract feature extraction, DCs often struggle to explain these misclassifications (the "black box" problem). In contrast, generative classifiers (GC) offer more transparent feature extraction and are less prone to misclassifying OoD data.

In this research project for the course Advanced Deep Learning in Robotics, we aim to build on Tulbure and Bäuml's work [1] by further analyzing the dataset and implementing a generative classifier similar to that in [2]. This approach could enhance classification robustness towards new materials and improve explainability by addressing the "black box" limitations of discriminative classifiers while maintaining competitive accuracy. This could enable robotic systems in real world scenarios to give a measure of confidence into their material classification, thus making it possible to utilize them in less restrained environments.

II. Related Work

Tulbure and Bäuml [1] extended the TactNet architecture, a convolutional neural network for tactile sensing, with state-of-the-art training protocols, achieving improved classification accuracy. To benchmark this model against human performance, they developed an experimental setup using tactile input alone, where their model consistently outperformed human classification.

They also released a dataset [3] of 36 common household materials, collected with the humanoid robot Agile Justin in a real robotic environment. Unlike controlled test bench setups common in the field [4], this dataset reflects natural variability and noise, improving transferability to real-world conditions.

Mackowiak and Ardizzone [2] explored generative classifiers as an alternative to discriminative classifiers in image classification. Using the Information Bottleneck [5] principle, they maximize information on the target class in the latent space while minimizing irrelevant input information, reducing overfitting. Because invertible neural networks (INNs) used as generative classifiers retain all input data, dequantization adds noise to enable the information bottleneck principle. Their generative classifier improves upon standard discriminative classifiers in explainability and OoD detection but does not offer added robustness to adversarial attacks.

III. Experiments

For our experiments, we first reimplemented the TactNet-II in a more simplified version, to have an own benchmark, which we can compare our invertible model architectures against. We stripped the original TactNet-II of the Siamese differentiation network architecture. Additionally, instead of a targeted adversarial training, we augmented the training data with random noise to train a robust model, whilst simultaneously decreasing training complexity. Apart from these changes, we utilized the same architecture as described in the paper, and trained with the same hyperparameters [1].

After setting up and training our TactNet-II model, we explored how the sequence length of the samples influences classification accuracy. Goal of this was to search for the saturation point, after which the model does not perform any better, even if it is given longer samples. This could be helpful for real world applications, when not always a defined sequence length may be given, and we could define a threshold, before which we are not able to classify. Additionally, it could make it easier to generate and process training data, as shorter sequence lengths also mean less work to prepare the samples. For this, we adapted the convolutional layers of the original architecture to also support shorter sample sequence length ranging from 100 timesteps to the original 1000 timesteps. We then defined a window with a set length (e.g. 100 timesteps), and then for each sample selected a single shortened sample randomly from within the sample. We trained the model with 10 different sequence lengths and validated with the same sequence length.

We have 1000x4x4 shaped data, as we record 1000 timesteps for each of the taxels on the 4x4 matrix. We collapsed the two spatial dimensions and separated the taxels, to simplify the architecture of the network. We then used the real Fourier transform operation on our single-taxel samples.

*Note: equal contribution

From the Fourier transform, we receive a real and imaginary part, both with the respective sequence length of 501. We then concatenated these two parts to a single sample with the length 1002. We then proceeded to batch the 16 taxels of each original sample together, to be processed by the network. We then implemented a simplified version of the IBINN classifier as described in [2]. Instead of using the FrEIA *Affine Coupling Block*, we resorted to the *All In One Block* and modified it, so that the permutation always swaps the first half of the data with the second. This block takes care of constructing the coupling block, the permutation, and the normalization, given a subnetwork architecture. For this subnetwork architecture, we relied on a simple fully connected subnetwork composed of two linear transformations separated by a ReLU activation. We set the hidden layer size to 512, with the in- and output dimensions being 1002, to be able to process our Fourier transformed data samples. Additionally we initialized a Gaussian Mixture model with a fixed variance and learnable parameters μ_k for k clusters, which represent the number of materials in our dataset. We initialized the μ randomly. Our information bottleneck loss is an almost identical version of the loss described in [2], and is the total loss in equation 3.

$$\mathcal{L}_{\text{gen}} = \frac{1}{D}\left[-\log\sum_{k=1}^{K}\exp\left(-\frac{1}{2}d_k\right) - \log|\det J|\right], \quad (1)$$

$$\mathcal{L}_{\text{cls}} = \text{CE}\left(-\frac{1}{2}\,d,\, y\right), \quad (2)$$

$$\mathcal{L} = \mathcal{L}_{\text{gen}} + \beta\,\mathcal{L}_{\text{cls}} \quad (3)$$

L_{gen} computes the normalized negative log-likelihood of the squared euclidean cluster distances between the latent space and the cluster centers of the Gaussian Mixture Model, encouraging the model to capture the data distribution accurately. It also includes a regularization term via the determinant of the Jacobian, ensuring stable transformations. L_{cls} on the contrary applies the cross-entropy function on the logits and the true labels y, to compare the model's predictions with the true labels. This loss steers the network to generate discriminative features that improve classification accuracy. The total loss function is acquired through the combination of these two losses, balanced with the hyperparameter β, which defines how the two are weighted according to each other.

IV. Results

1) TactNet-II Architecture Replication: Our simplified model achieves 77.2 % accuracy, which, although lower than the benchmark of 86.3 %, serves as a solid baseline for evaluating other architectures. The model trains very smoothly and shows a clean loss curve for both training and validation loss (Fig. 1. The model converges after approximately 40 epochs.

It can be seen in Fig.2 that our network performs well on classes that the original TactNet-II also performs well on, and struggles with the classes the original also performs poorly on.

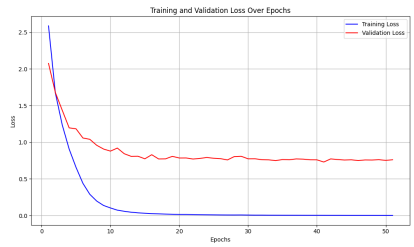

Fig. 1. Training and Validation Loss across Epochs

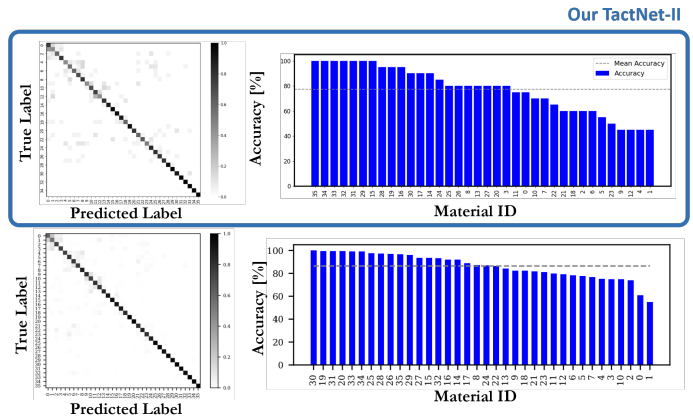

Fig. 2. Our Implementation vs. Original TactNet-II

The discrepancy in the accuracy can be assumed to be caused by the simplifications of our network architecture compared to the original implementation. It is especially notable that both models on average perform more poorly on materials with a low index number. This could be due to the nature of the materials, or due to how the dataset has been recorded. A future experiment should focus on what makes these materials different, by comparing the samples, investigating patterns on what materials they are misclassified as and looking into the data generation of the dataset.

2) TactNet-II Sequence Length Sensitivity Study: As can be seen in Fig.3 the sequence length did not show a saturation point at which further prolongation resulted in higher accuracy. This could indicate that the saturation point lies above the maximal sequence length of 1000 time steps and that a dataset with longer timesteps would yield even higher classification accuracy.

However, it is important to note that the random selection of our window from within the sample could also exclude essential timesteps that greatly influence classification performance. If the first contact of the sensors with the material holds extremely high importance for classification, the linearity in our plot could be explained by how often this critical point is included within the window. Therefore, a larger window would result in a higher probability of including this essential point, thus leading to greater accuracy. As we linearly increase this probability by linearly increasing the window size, the accuracy also increases linearly. A subsequent experiment

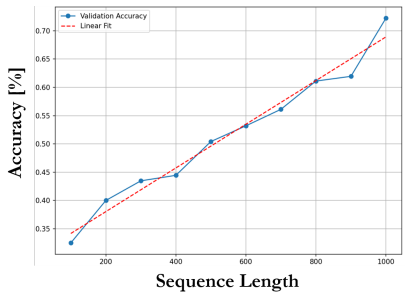

Fig. 3. Accuracy over Sequence Length

should focus on isolating whether such essential points exist, and if so, how the sequence length could be reduced while still including these datapoints. If no such singular points exist, an experiment could be conducted to investigate how far this linearity scales, potentially raising the benchmark of the TactNet-II classification accuracy.

3) Implementation of TactINN: The TactINN achieves a classification accuracy of 32.8 % (Fig.5). Although this is lower than the current state-of-the-art, the model appears to learn semantic features from the data, as illustrated by the characteristic diagonal in the confusion matrix.

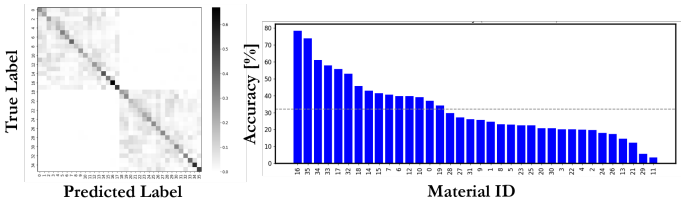

Fig. 4. Performance of TactINN

After conducting a hyperparameter search, we recieved the following best performing parameters.

Parameter	Values
Learning Rate	1×10^{-3}
β	3
Batch Size	16
Optimizer	ADAM
n Coupling Blocks	4
Weight decay	$1e^{-4}$
Hidden Layer Size	512

Fig. 5. Hyperparameters for TactINN

When visualizing the latent space distribution of the validation samples after performing PCA to reduce the latent dimension to 50, and then TSNE to further reduce them to 3 dimensions, we can see that the samples group together to some extent to form clusters. This means that there is a similarity between the samples that is recognized by the model and appropriately represented in the latent space.

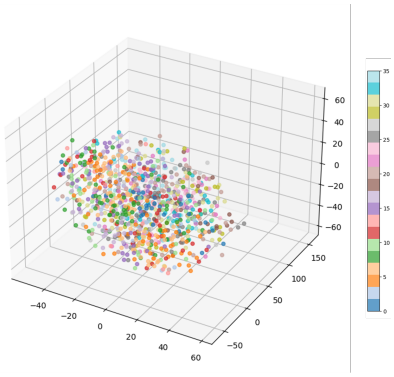

Fig. 6. Low Dimensional (3D) GMM Visualization of the Latent Space

This confusion matrix shows an exact split between the first and second 18 classes, suggesting that the GMM is grouping the data into two hyper-clusters rather than 36 distinct clusters, which is also reflected in the low-dimensional latent space representation (Fig.7). The perfect division is unusual, as the model consistently separates the data without misclassifying material classes between hyper-clusters, even though the original TactNet-II confused them. This could result from differences in data recording—for example, separate sessions introducing bias that is evident in the Fourier-transformed data which is used in the TactINN but not in the original samples that are used in the TactNet-II.

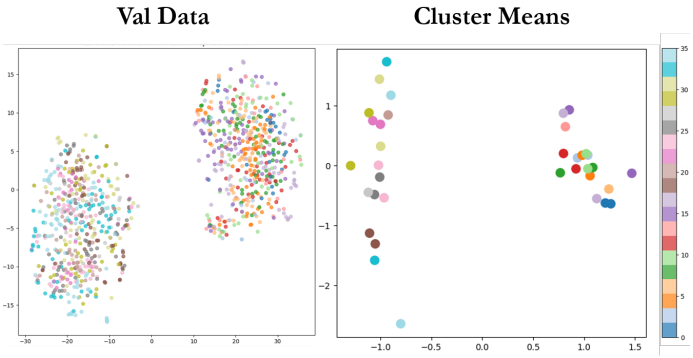

Fig. 7. Low Dimensional (2D) GMM Visualization of the Latent Space

Another possibility for the collapse of the GMM is the initialization of the 36 cluster μ. If these are initialized too closely or with an inherent structure, a clear differentiation may be induced early in training. We tested various initializations—including data-based initialization via GMM pretraining, simple zero tensors, and a scaled identity matrix as implemented in the original paper. Although random initialization performed slightly better in our tests, no significant differences were observed, and the GMM consistently forms two hyperclusters. This suggests that the issue, if not originating from the data, may lie elsewhere in the model rather than in the GMM. The cause also could be a bias introduced by the model architecture itself, which might include operations that semantically separate the first and second 18 classes. Further

analysis should focus on whether the split is induced by the model or the data; for instance, by switching the labels to see if the perfect split persists, and by analyzing the Fourier-transformed data for frequency patterns that differ between the two halves. If the formation of the hyper-clusters is data based, future preprocessing steps of the samples could include operations emphasizing the differences between all classes, whilst normalizing the differences between the two hyper-clusters. This would ensure the model focuses more on the differences between the classes in general.

Finally, future work could include sampling of the latent distribution, more specifically distinct clusters and comparing it to the original data samples of the corresponding class. This would consolidate that the model understand the structure of the data and what makes the materials differ from each other.

V. Conclusion

In conclusion, our study builds upon the work of Tulbure and Bäuml [1] and Mackowiak and Ardizzone [2] by exploring both discriminative and generative classification approaches for tactile sensing. Our simplified replication of TactNet-II achieved competitive performance while shining new light on sequence length and its affect on accuracy. The sequence length sensitivity study indicates that longer sequences might further improve classification accuracy, while also suggesting that there exist a number of timesteps that play an important role in model performance. Future work could further investigate the existence and effect of these essential timesteps, and propose how a minimal viable sample could be defined, to ensure a quality threshold for tactile sensing samples in real world scenarios.

Furthermore, we implemented invertible neural networks and generative classifiers in a novel way and showed that generative classifiers can be used for material classification. Our implementation of the TactINN, although underperforming the benchmark TactNet-II model, achieved an accuracy of 32.8 %. Further analysis of the confusion matrix and low dimensional representation of the latent space revealed an interesting collapse of the Gaussian Mixture Model into two hyper-clusters rather than the expected 36 distinct clusters. This raises questions about potential biases either in the data or the model architecture. Future work should focus on isolating these factors, potentially through label switching experiments and a deeper analysis of the Fourier-transformed data, to determine whether the split is induced by the model or the data.

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